

MISM 6405: Data Wrangling for Business



ESCAPE ROUTES

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September 19, 2025

Data Source



Where does our data come from?



- **Source:** Kaggle – Airbnb Open Listings dataset, widely used for hospitality and travel research
- **Dataset Size:** 74,111 listings with 29 variables
- **Variables include:**
 - Property Information: Property type, room type
 - Location Data: Latitude, longitude, neighbourhood, and zip code details for spatial analysis
 - Pricing & Demand: Listing price per night, availability calendar, minimum nights requirement
 - Customer Interaction: Number of reviews, review scores, ratings that reflect guest satisfaction
 - Host Characteristics: Host name, verification status, superhost badge, response rate, and profile details
 - Amenities Offered: Wi-Fi, kitchen, parking, pool, air conditioning, and other features influencing price and demand



Background of the Data

UNDERSTANDING THE DATASET

About Airbnb

Founded in 2008, Airbnb has grown into a global leader in the peer-to-peer lodging market, transforming hospitality and travel worldwide

Dataset Coverage

Includes over 74,000 listings across multiple U.S. cities
Captures detailed information on property types, neighborhoods, and host characteristics
Provides both quantitative data (price, availability, reviews) and qualitative details (amenities, descriptions)

Business Relevance

Reflects real-world consumer behavior in the hospitality sector
Provides insight into supply (hosts, properties) and demand (reviews, bookings)

Project Objective



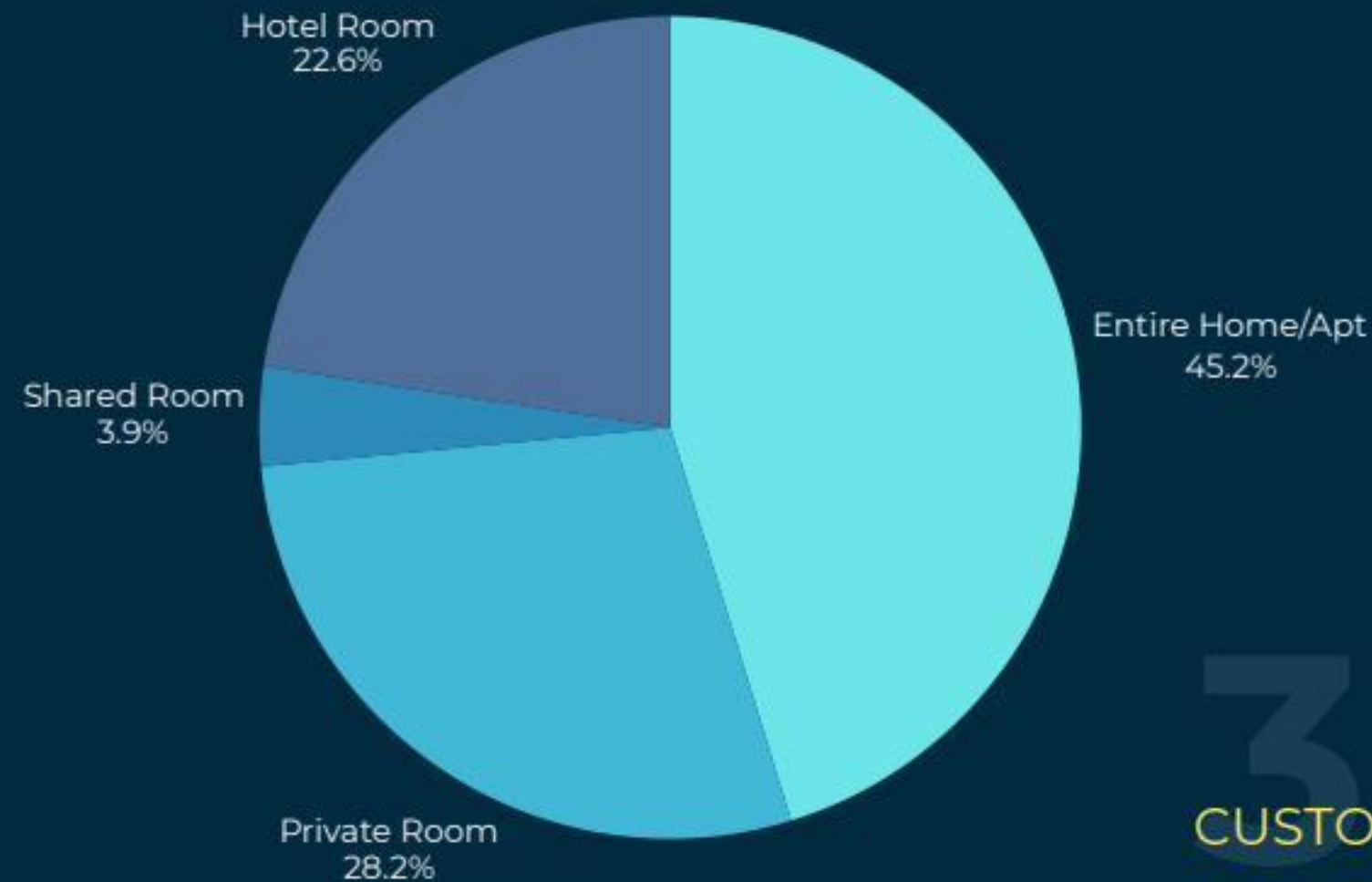
Our main goal

1 PRICING PATTERNS

Compare average prices across cities and cancellation policies. Identify which locations command higher prices and why.

2 GUEST CAPACITY

Examine how different room types affect the number of guests accommodated. Highlight the alignment of room types with traveler groups.



3 CUSTOMER SATISFACTION

Measure review score ratings across property types. Explore the relationship between number of reviews and rating reliability.

4 BUSINESS STRATEGIES

Provide strategies for hosts to optimize pricing and listing features. Recommend improvements to Airbnb for enhancing competitiveness in the market.

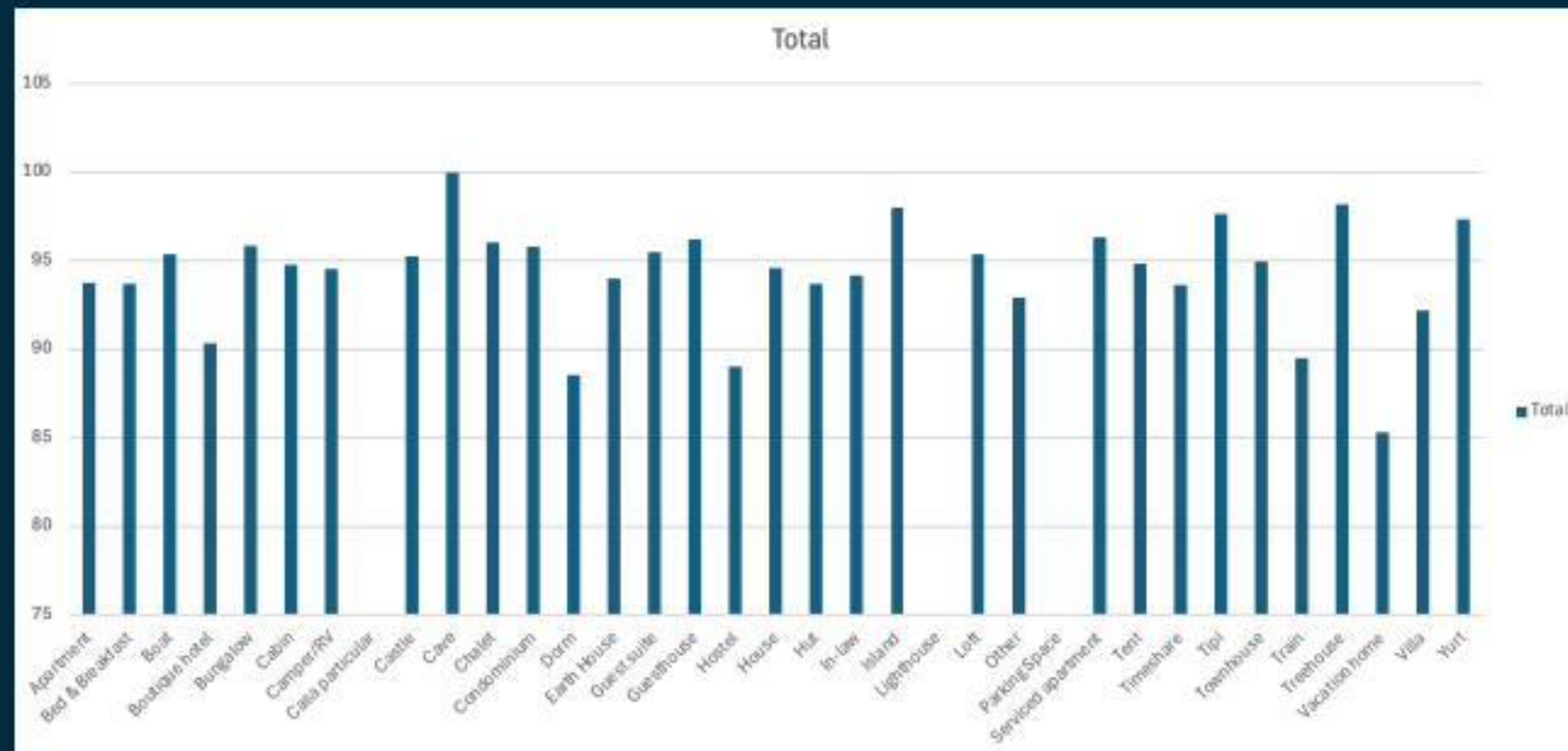
Business Problems to Explore



Key business questions

HOW

do average Airbnb prices differ across cities?



WHICH

property types need improvement based on lower review scores?

Business Problems to Explore



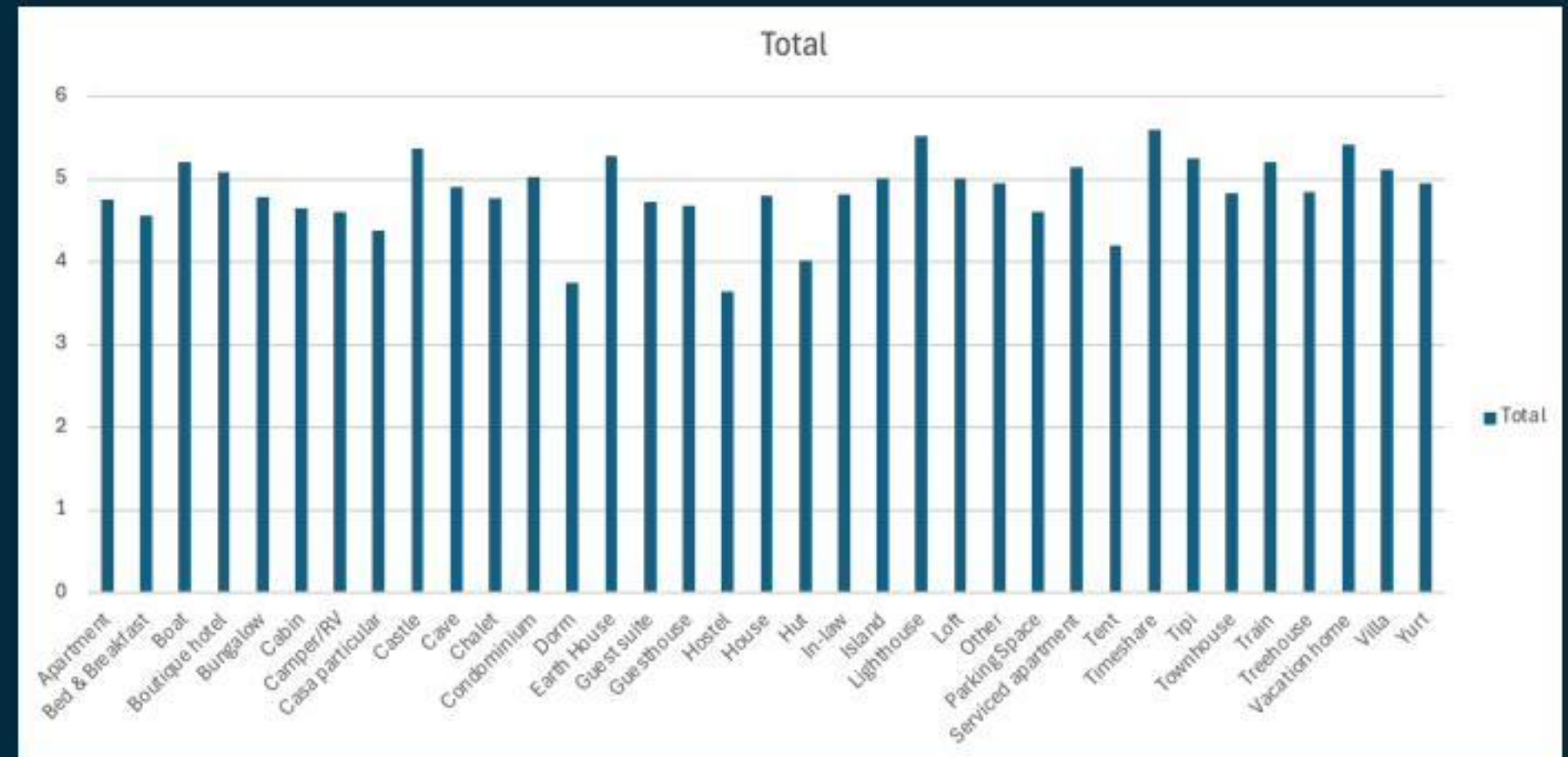
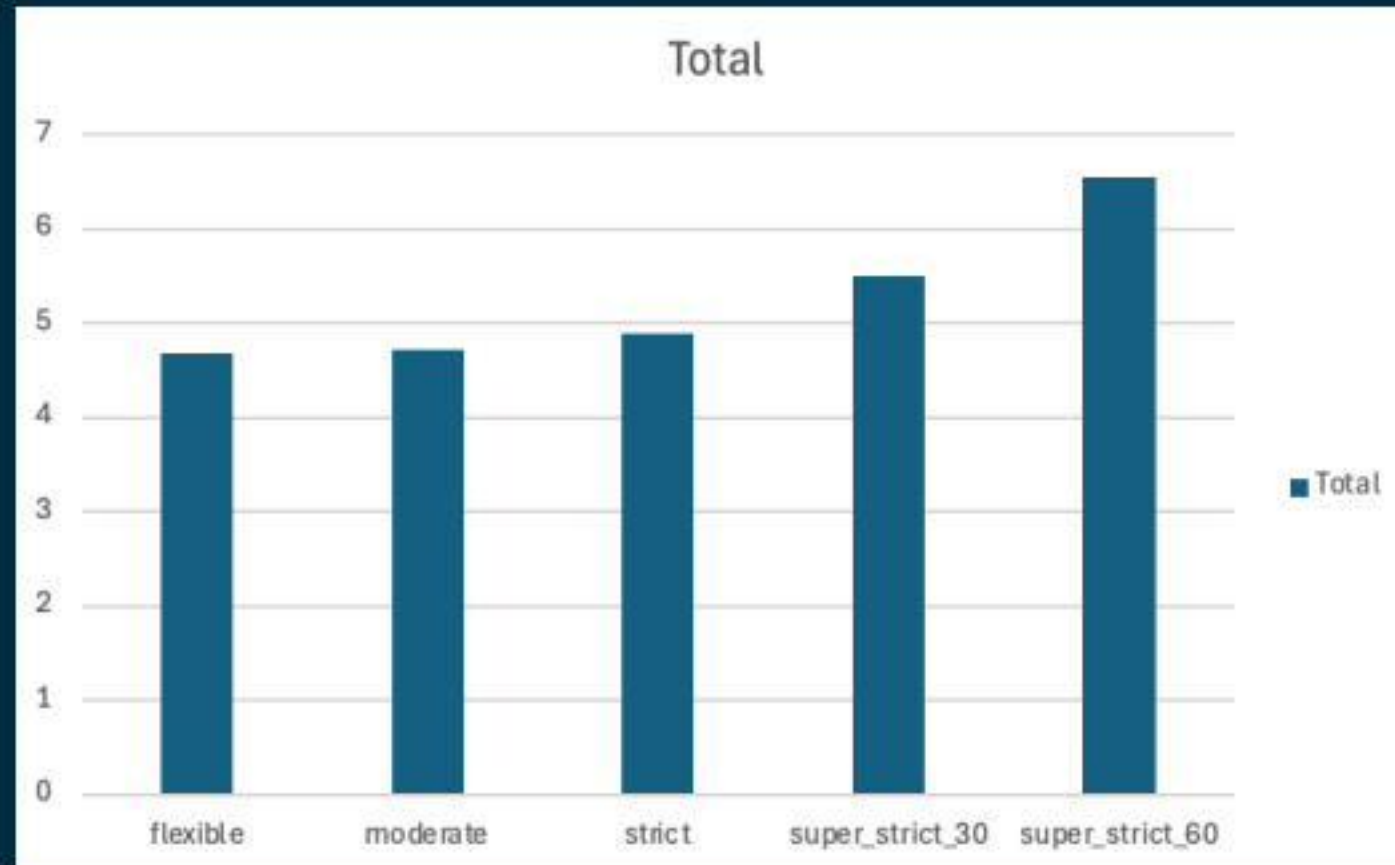
Key business questions

WHAT

is the impact of stricter cancellation policies on listing prices?

WHICH

property types are the most affordable vs. most expensive?



Research Method



How we plan to analyze the data

DATA WRANGLING



Clean and standardize the dataset by fixing missing values, duplicates, and inconsistent formats.

EXPLORATORY DATA ANALYSIS



Identify patterns and trends in pricing, reviews, and neighborhood demand.

OPTIMIZATION



Identify the best pricing or amenity strategies for hosts.

VISUALIZATION



Build dashboards in Tableau to present insights in an interactive and business-friendly way.

Potential Results



Pricing Insights

- Identify the main factors that drive listing prices, such as room type, neighborhood, and amenities offered
- Highlight patterns in overpricing or underpricing across different property types
- Provide benchmarks for competitive pricing strategies



What insights can we expect?



Guest Preferences

- Determine which room types are most popular among travelers
- Analyze which amenities are most strongly associated with higher bookings



Host Profiles

- Discover characteristics of successful hosts
- Show how host engagement and responsiveness influence reviews and bookings



Market Segmentation

- Create a timeline with milestones
- And responsibilities. Set KPIs to measure success.

Practical / Managerial Implications



HOW THE RESULTS CAN BE APPLIED



FOR HOSTS

- Set competitive nightly rates based on neighborhood and property type
- Identify which amenities to add or highlight to attract more bookings
- Improve host profiles (verification, fast responses) to gain customer trust



FOR AIRBNB PLATFORM

- Enhance recommendation systems with better price and demand insights
- Improve search filters by highlighting high-demand amenities or neighborhoods
- Strengthen trust signals through host verification and review analysis



FOR CITIES & TOURISM BOARDS

- Understand demand hotspots to guide tourism infrastructure planning
- Track the impact of Airbnb on housing markets and local communities
- Support data-driven tourism policies and regulations



FOR CUSTOMERS

- Find listings that balance price, amenities, and location for better value
- Gain transparency on what factors truly affect pricing
- Use reviews and host information to make safer and more reliable booking choices



Thank You for your attention

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